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Design optimization of three dimensional geometry of wind tunnel contraction

A.S. Abdelhamed, Y.El-S. Yassen *, M.M. ElSakka

Mech. Eng. Dept., Faculty of Eng., Port-Said Univ., Port-Said, Egypt

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KEYWORDS

Wind tunnel contraction;
Turbulence model;
Goal-driven optimization;
Screening method;
Uniformity;
Analytical technique

Abstract The aim of the present study is to redesign three dimensional geometry of existing open circuit wind tunnel contraction. The present work achieves the recommended contraction ratio, maximum uniformity at the working section mid-plane, without separation, no Gortler vortices in the contraction, and minimizing the boundary layer thickness at entrance to the working section. Using CFD along with optimization tools can shorten the design optimization cycle time. Moreover CFD allows insight into the minute flow details which otherwise are not captured using flow bench tests. The design exploration algorithm is used to optimize the profile of the contraction in an automated manner. The optimization is based on using screening method to choose the best design set and verified by the CFD solver. The new contraction, compared to the old design contraction is confirmed using CFD. The new design is manufactured in full scale. The optimized contraction is investigated computationally and experimentally.

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1. Introduction

Contraction is an important part of a wind tunnel. The main effects of a contraction are to reduce both mean and fluctuating velocity variations to a smaller fraction of the average velocity and further to increase the corresponding mean velocity. Generally to design of a subsonic and supersonic wind tunnel, the contraction portion should not have adverse pressure gradient in the stream-wise and further the effect of adverse pressure gradient at the exit of the contraction must be minimal. Whenever a converging duct segment is attached to constant-area segments, regions of adverse pressure gradient will occur along the wall, at its inlet and exit that may cause boundary layer separation. If separation occurs, it will degrade the flow uniformity and steadiness, both of which are essential in a test facility. Separation is usually avoided if the adverse pressure gradients are minimized which is done by making the contraction sufficiently long. The contraction can be divided into two parts. The first part has walls of concave shape and it is very important to elongate this part as much as possible to avoid wall boundary layer separation. The streamline curvature effects on the pressure gradient in the boundary layer promote the risk of separation. Along a fair part of this section, there will be a positive pressure gradient. The second part of the contraction

* Corresponding author. Tel.: +20 10 1246 1182.
E-mail addresses: sh_ahmed99@yahoo.com (A.S. Abdelhamed), y_yassen70@yahoo.com (Y.El-S. Yassen), elsakka_mohamed@yahoo.com (M.M. ElSakka).

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Nomenclature

a	concave-straight line deviation relative to the contraction inlet half height, h
b	convex-straight line deviation relative to the contraction inlet half height, h
c	axial position of the inflection point relative to the contraction length, L
D_h	test section hydraulic diameter
Uni_{ex}	flow uniformity at the exit plane of the contraction
τ_{min}	minimum wall shear on the contraction, N/m^2
β	porosity
C_2	pressure jump coefficient
k	pressure drop coefficient

has convex walls that may cause flow separation in the vicinity of the test section due to a positive pressure gradient. To delay separation, it is better that a longer contraction's length be chosen, but this will increase the cost and thickening the boundary layer that may enhance boundary layer and risk of separation. Furthermore, if the length is reduced, the contraction costs will reduce and it will fit into a smaller space. In addition, the boundary layer will generally be thinner due to the combined effects of increase in the favorable pressure gradients and decrease in the length of the contraction. Furthermore, it may increase the possibility of flow separation. Thus the length most be optimized. The contraction area ratio is another dominant factor that affects the extent of flow uniformity, flow separation, and downstream turbulence level. The characteristics of the flow were investigated in a square contraction numerically and experimentally [1]. Their Measurements included the cross-sectional velocity profiles and longitudinal pressure distributions along the wall of a contraction of a wind tunnel. Boundary layer was studied in the contraction [2]. The calculations showed that the minimum-length contraction shapes can provide fully attached boundary-layer flow. The results showed that exist regions of separated flow along the wall, in the inlet and exit of the contraction furthermore; conclusions were drawn that the existing adverse pressure gradient is the essential condition for the separation. Flow study shows that a three-dimensional separation occurs in the contraction surface. In a proposed conceptual model of this phenomenon, the separation process begins with small non-uniformities in the boundary-layer flow merging from the screens upstream of the contraction. On entering the contraction, the non-uniformities are amplified by a combination of Gortler instability, lateral pressure gradient and adverse streamwise pressure gradient to form a strong counter-rotating streamwise vortex pair (Gortler vortex) that detaches from the surface [3,4].

Another contraction parameter that has to be selected, a priori, is the cross-sectional shape. In order to avoid cross-flows and boundary layer separation in the corners, the ideal cross-sectional shape is circular. However, in the absence of separation, the secondary flows in the corners tend to remain localized, without any significant effect on the test section flow quality [5]. The cross-sectional shape for modern day contractions is, therefore, almost always chosen to match the other tunnel components which are normally square or rectangular.

An iterative design procedure was developed for the contraction to be installed on the mixing layer wind tunnel [6,7]. The procedure consisted of first computing the potential flow field and hence the pressure distributions along the walls of a contraction of given size and shape using a three-dimensional numerical panel method.

Currently, more flexibility in the design of wind tunnel contractions can be exhibited, with the use of CFD to enable rapid testing of designs to optimize contractions of arbitrary cross-section and wall profile. The use of CFD allows for the use of design exploration algorithm to optimize the profile of the contraction in an automated manner. However, the performance of the contraction still requires testing after construction, as the level of CFD used for this application is typically insufficient to detect the development of longitudinal vortices through the working section such as were measured by [8].

A great effort has been made to set the guidelines for the design of wind tunnels [9,10]. However, the recent development in optimization techniques encourages the researchers to develop optimized design for the wind tunnels especially the wind tunnel contraction that has a strong influence on the flow quality in the test section. An optimization is performed for a 2D contraction profile described by a six order polynomial and the results are validated experimentally [11]. The effective Global Optimization algorithm is used to optimize the profile of a 2D contraction described by a two parameters Bézier curve modeled by a three- dimensional potential flow solver using Kriging Meta model to predict the values of the objective function [12].

The present work contains a detailed account of the redesign of the existing contraction that based on the analytical technique according to the method given by [13,14], assembly and calibration of a wind tunnel specifically redesigned for maximum flow uniformity at the working section, prevention of separation and no Gortler vortices in the contraction, controlling the turbulence level and minimizing the boundary layer thickness at entrance to the working section. The wind tunnel consists of two separate legs which are driven independently by centrifugal blowers connected to variable speed motors. The two streams are allowed to merge in the test section. The construction of the wind tunnel was motivated by a strong interest in the study of three dimensional contraction profiles. The wall contraction profile is described by three dimensional delineated curves and optimized using the design exploration algorithm based on screening method. The flow in the wind tunnel is modeled using a Reynolds-averaged Navier–Stokes (RANS) solver. The aerodynamic performance improvement of the optimized contraction, compared to the old design contraction based on analytical technique, is confirmed using CFD. The new design is manufactured in full scale. Although, investigation by using CFD analysis is less expensive in general, the results obtained from CFD calculations should be validated by means of experimental results. In addition to validation, in cases like simulating three dimensional contraction profiles, performing CFD simulations can

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be time consuming. Therefore, it is essential to find approaches that can reduce the computational cost while results are validated by experiments. The performance of the optimized contraction is investigated computationally and experimentally.

2. Wind tunnel description

The wind tunnel facility included in this work is an open jet blower type driven by two similar 5.5 kW centrifugal blowers. Each blower supplies air into a separated branch through a flexible joint. Each branch consists of two wide angle diffuser connected in series. The first diffuser section expands from 36×50 cm to 50×100 cm over a length of 125 cm, giving a maximum angle of 22.6° and an area ratio of 2.78. The second diffuser expands from 50×100 cm to 74×150 cm over a length of 100 cm, giving a maximum angle of 28° and an area ratio of 2.22. A screen with a porosity of 0.43 is placed between the two diffusers while a screen with a porosity of 0.5 is placed at the exit of the second diffuser. The flow from the two branches is merged in the settling chamber. The merged flow passes through the contraction section as shown in Fig. 1.

The old contraction has a square cross section of 150×150 cm^2 at inlet, 80×80 cm^2 at exit, and a contraction ratio (CR) of 3.5 that is lower than the recommended contraction ratio (6:9) [13,14]. The optimized contraction has an exit cross section of 60×60 cm^2 and a contraction ratio of 6.25 over a length (L) of 150 cm that gives the recommended ratio of unity between length and hydraulic diameter at the contraction inlet [14]. The two geometries of old and optimized contractions are shown in Fig. 2.

3. Experimental procedure

The geometry of optimized contraction was manufactured in full scale. In order to achieve the inlet boundary condition (12 m/s) compared to the CFD calculation, the two similar 5.5 kW centrifugal blowers are to be controlled by an electrical inverter. The velocity profile at exit section of the contraction is measured using Static-Pitot tube of 6 mm diameter. A simple slide probe traversing mechanism is used to locate the probe tip at every location (30 points, and 1 cm between each two successive points and three points, 2 mm between each near the wall except the nearest to which is at 4 mm from the wall). The collected measurements data at measuring station locations were averaged and used to calculate the location mean velocity distribution of the air flow through the duct, for Reynolds number ($Re = 4.8 \times 10^5$). The uncertainty, using root-sum-square method in the measured mean velocity is $\pm 4.8\%$ within 95% confidence.

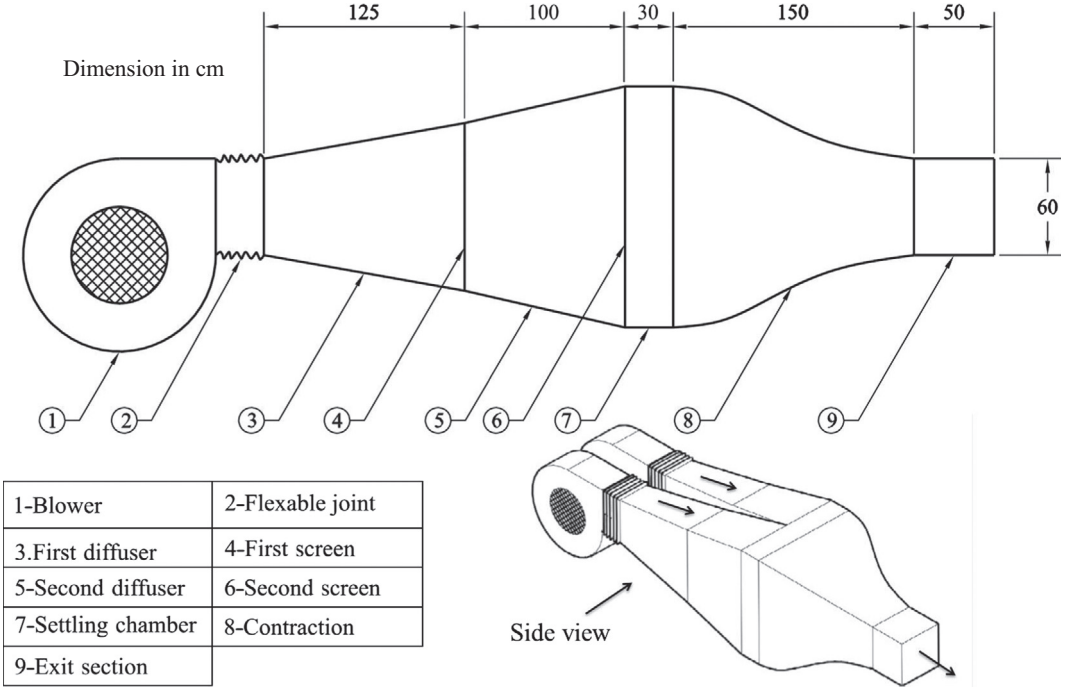
4. Design exploration and optimization

The optimization process is performed using the Design Exploration algorithm described in Fig. 3. The optimization is the process to choose the optimal input parameters that achieve the desired performance of the system in an automated manner. This performance is evaluated by means of output parameters. There are several choices to examine the system performance using a certain output parameter such as maximize its value, minimize its value, or seeking target value.

Initially, the complete analysis of an initial design has to be held including drawing, meshing, solving the governing equations, defining the input and output parameters.

4.1. Problem parameterization

The profile of the wind tunnel contraction consists of concave and convex parts. The method of the delineation of the profile is based on the perception of its variation from a straight line [15]. This method has been utilized to parameterize the profile



1-Blower	2-Flexable joint
3.First diffuser	4-First screen
5-Second diffuser	6-Second screen
7-Settling chamber	8-Contraction
9-Exit section	

Figure 1 Wind tunnel.

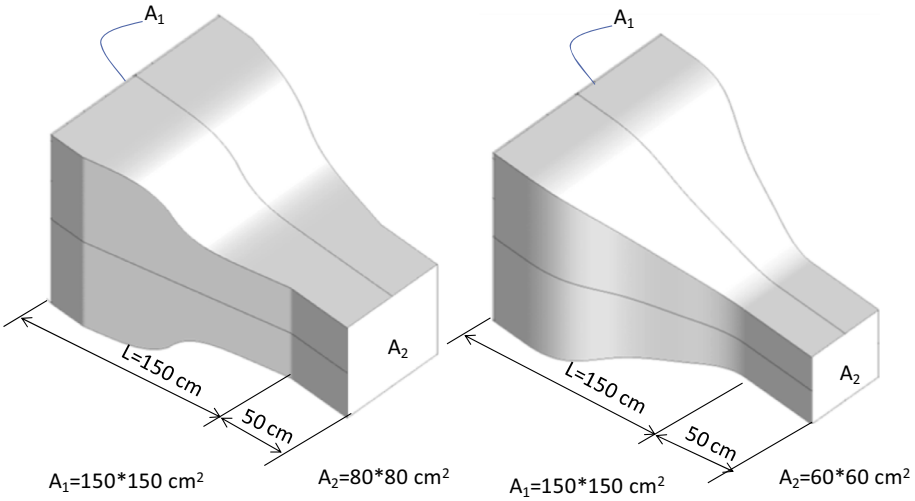


Figure 2 Shape of the old contraction (left) and the new design (right).

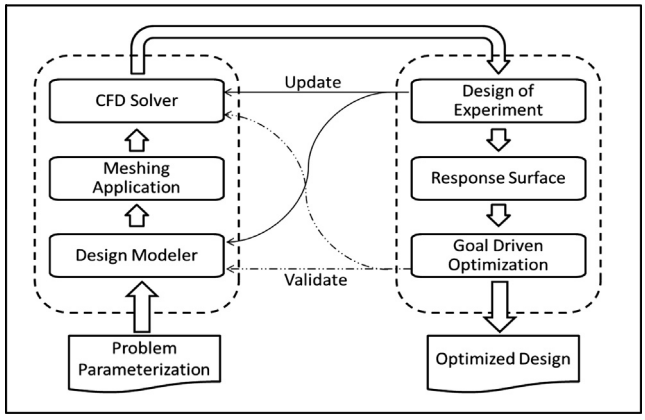


Figure 3 Design exploration algorithm.

of the contraction into three dimensionless parameters *a*, *b*, and *c*. These parameters are defined in the Design Modeler to describe the profile and the variation of the profile of the contraction as shown in Figs. 4 and 5. The objective function of the optimization process is to maximize the flow uniformity at the exit plane of the contraction, Uni_{ex} where:

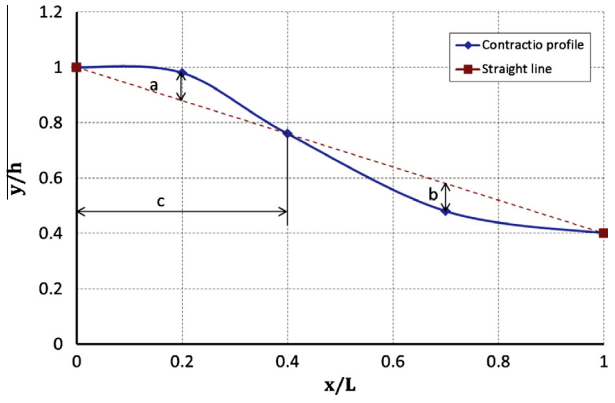


Figure 4 The contraction profile parameterization.

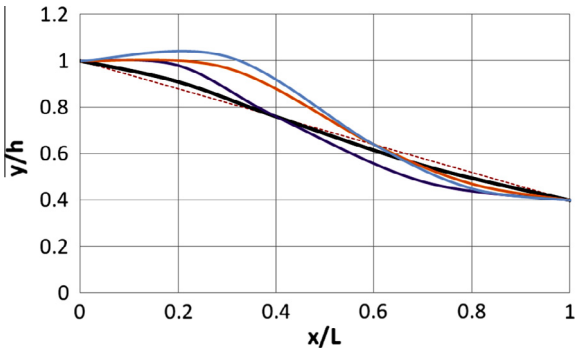


Figure 5 The variation of the profile with dimensionless parameters *a*, *b*, *c*.

$$Uni_{ex} = \left|\frac{U_{max}}{U_{mean}}\right|_{exit\ plane} \quad (1)$$

The minimum positive wall shear on the contraction is used to attribute the absence of separation in the contraction.

4.2. Computational treatment

The CFD solver used for this study is a Fluent (ANSYS 14) Pressure Based Segregated Solver. Quadratic Upwind Interpolation

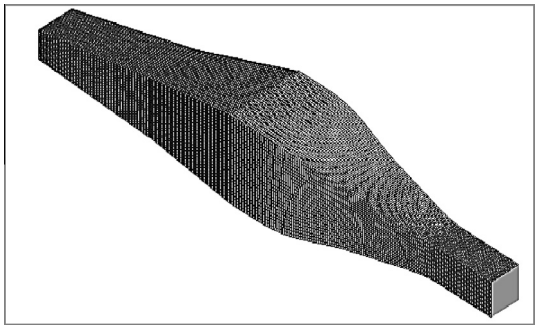


Figure 6 Hexahedral mesh of the final design one quarter of the wind tunnel (grid representation).

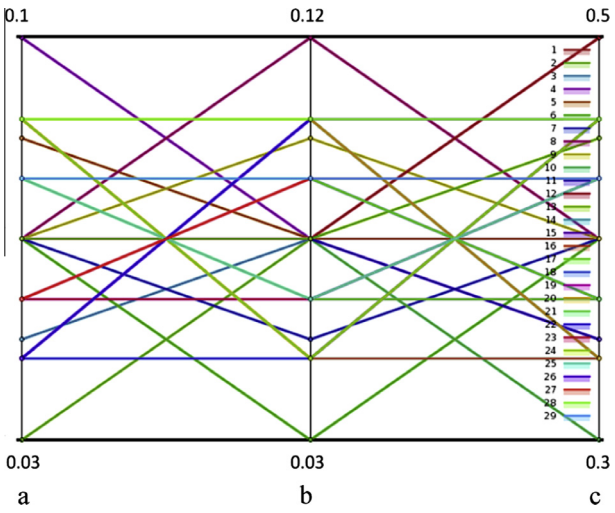


Figure 7 Parameters parallel chart for 29 sample points with input parameters.

(QUICK) that is a 3rd-order accurate is used for spatial discretization. Shear Stress Transport (SST – k–ω) model is used for turbulence modeling. One quarter of the wind tunnel is modeled, using the horizontal and vertical symmetry planes to reduce computational time. The domain is meshed using a hexahedral mesh. The mesh of the final design shown in Fig. 6 has 254,259 elements with a maximum aspect ratio of 2.7778 and average skewness factor of 0.089395. The extensive numerical simulation illustrates that the hexahedral mesh is more stable and robust to converge. In spite of good results, usage of this type makes the mesh not fine enough near the wall to resolve the characteristics of the boundary layer. The inlet boundary condition to the domain is 12 m/s. The outlet boundary condition is assumed to be constant static atmospheric pressure at exit.

The screens are represented using the porous jump model with a pressure jump coefficient of $C_2 = 218.875$ [1/m] for the first screen and 142 [1/m] for the second one. These coefficients are derived by the relation provided by [16,17] for the pressure drop coefficient $k = 0.71 * (1 - \beta)/\beta^2$.

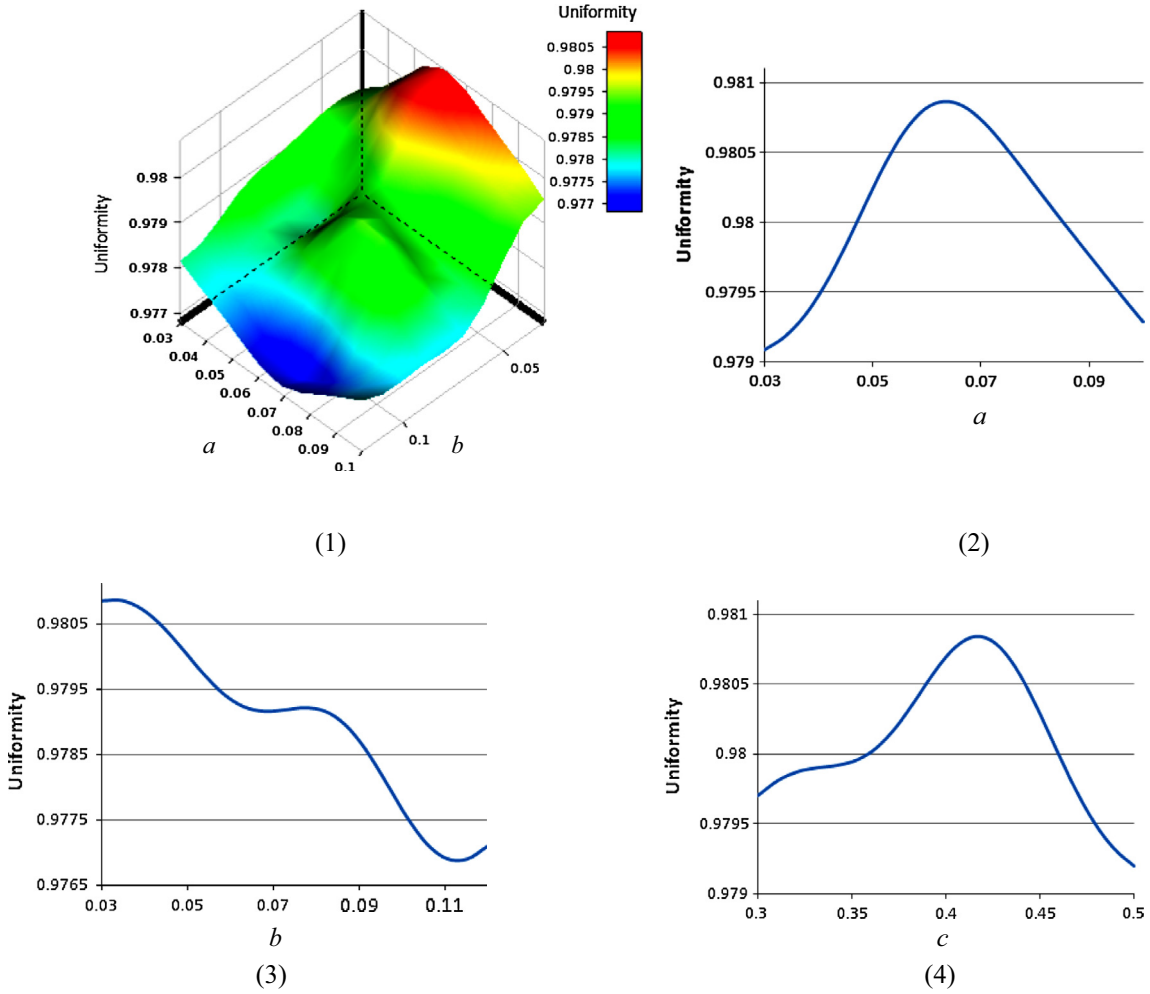


Figure 8 (1) Effect of both a and b on the uniformity, (2) effect of a on the uniformity, (3) effect of b on the uniformity and (4) effect of c on the uniformity.

Table 1 Optimization design candidates and its verifications.

Candidate points	*a*	*b*	*c*	Uni_{ex}	Minimum wall shear
Candidate (A)	0.088835	0.036637	0.325706	0.981661211	0.003984247
Verification (A)				0.979834	0.00579331
Candidate (B)	0.087155	0.034527	0.34500	0.98129169	0.004430081
Verification (B)				0.979716	0.00658563
Candidate (C)	0.066435	0.035758	0.414229	0.980838786	0.010024012
Verification (C)				0.980373	0.00972408

4.3. Goal Driven Optimization

The Goal Driven Optimization is performed in three steps:

4.3.1. Design of Experiment (DOE)

In this step a test sample of the input parameters is generated to cover the whole range of these parameters with minimum number of design points, keeping this sample efficiently represents the effects of the input parameters on the output parameters. DOE technique tries to locate the sampling points in the design space such that the space for random input parameters is explored in most efficient way, or to obtain the required information with a minimum of sampling points [18].

After generation of the test sample in the DOE table using one of DOE methods. The update process is performed in which the optimizer repeats all the steps have been done in the initial design (drawing/meshing/solving generating output parameters) for all the design points in the DOE table.

4.3.2. Response surface

In this step the results generated in DOE table is utilized to build a multidimensional correlation. This correlation will be used to predict the response or the behavior of the system to any change of the input parameters. The predicted response surface is used to get the value of the output variables to any combination of the input parameters without the need to perform a complete simulation.

4.3.3. Optimization

In this step the predicted response surface is utilized to choose the optimal input parameters that achieve the desired performance or characteristics of the tested system using a comparative study by checking the output parameters to be maximized, minimized, seeking target value

In this study, the initial design has been setup. The Design of Experiments (DOE) technique is used for sampling. The Enhanced-Rotatable Central Composite Design (CCD) is

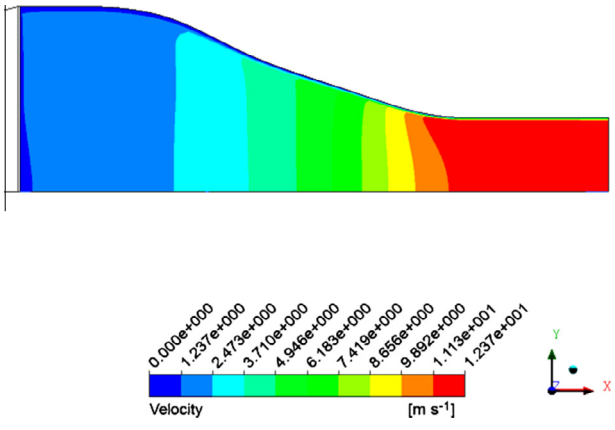


Figure 9 The velocity contour in the mid-plane of contraction.

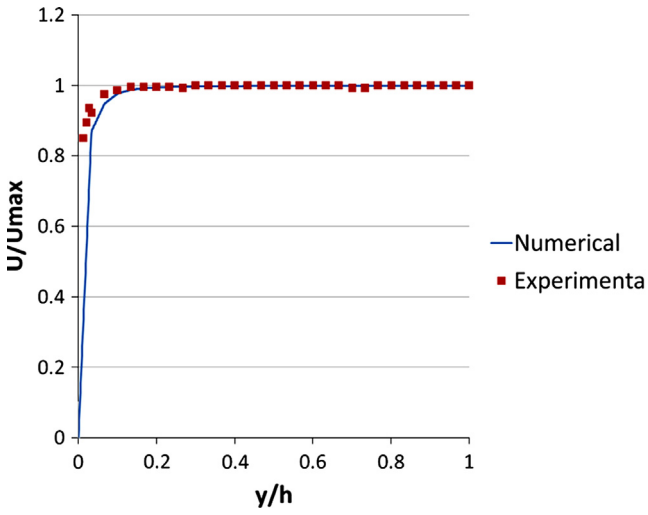


Figure 10 The velocity profiles at the mid exit plane.

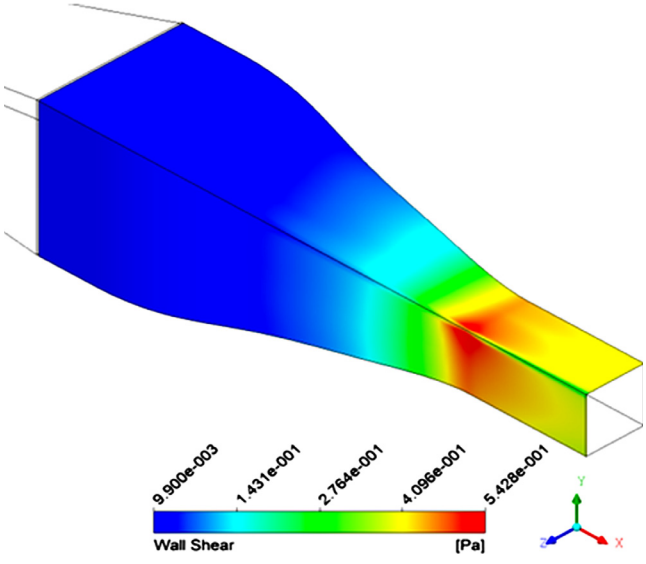


Figure 11 The contour of the wall shear of the contraction.

Table 2 Boundary layer characteristics at the exit of test section.

	Old	New	Experimental
Half hydraulic diameter $D_h/2$ (m)	0.4	0.3	0.3
Maximum velocity (m/s)	6.82286	12.1956	12.11892061
Dimensionless boundary layer thickness $2\delta/D_h$	0.181188	0.148143	0.11128
Boundary layer thickness δ (m)	0.072475	0.044443	0.033384
Displacement thickness δ^* using numerical integration (Trapezoidal method) (m)	0.009713	0.007492	0.002911251
The momentum thickness θ using numerical integration (Trapezoidal method) (m)	0.004984	0.003567	0.002338749
Reynolds number based on momentum thickness Re_θ	2168.764	2774.47	1807.596359

chosen for DOE that is a five level sampling design as the other three level sampling methods is insufficient to describe the problem. Fig. 7 shows the parameters parallel chart for 29 sample points with input parameters.

DOE sample points has sent to the CFD solver to update the value of the output parameter. These values are used to generate the response surface by fitting it using multidimensional regression techniques. In the current study, the Non-Parametric Regression (NPR) has been chosen as meta-modeling (surrogate) technique to create the response surface to meet the highly nonlinear behavior of the relation between the output parameter and input parameters.

Once the response surface has been created the so called Goal Driven Optimization (GDO) is performed based on screening method. The screening method allows generating a new sample set with 10,000 sample points and sort its sample based on the objective that is set to seek the maximum flow uniformity at the exit plane. The design candidates chosen by the optimizer are then verified using the CFD solver [19].

5. Results and discussion

Fig. 8 shows the generated response surface and the influence of the input parameters on the exit plane flow uniformity considering the optimal design set as a reference point. It is indicated that Fig. 8(1) shows the effect of both a and b on the uniformity. It can be observed that the uniformity is affected by changing parameter (a) as shown in Fig. 8(2). But, decreasing the trend of uniformity is attributed to increasing the parameter (b) as shown in Fig. 8(3). It can be seen from Fig. 8(4) that the maximum uniformity is occurred at 42% of the contraction length of the geometry profile, (i.e. inflection point) compared with the recommended ratio of 40%.

There is no guarantee for unique solution to have just one geometry but, there are some limitations should be considered to implement the design method. The optimization technique used has the capability to design the geometry. The design method gives three shapes (as shown in Table 1), all of them satisfy the requirements. The chosen geometry that can be validated experimentally and achieve the objectives. Table 1 shows the chosen design candidates and its verifications. It can be seen that candidate (C) has been chosen as the final design.

The computational model was reviewed to test for uniformity of flow in the working section. Fig. 9 shows the velocity contour in the mid-plane of the contraction. It can be seen that the flow is fully developed at the test section. Also, it can be observed that the contraction is free from separation and no tracing of Gortler vortices. However, the relation between the radius of the concave shape of contraction and the momentum thickness is not comparable, where the radius = 1 m, and the momentum thickness $\theta = 0.003567$ m computationally while $\theta = 0.002338749$ m experimentally. Also, the optimization method with certain constrain prevents adverse pressure gradient that causes separation in the contraction and non-uniformities that may transfer to the working section as shown in Figs. 9 and 10, respectively. In addition that there is a positive minimum shear stress at the wall of contraction as shown in Fig. 11.

The computational model demonstrated reasonably uniform flow across the wind tunnel mid plane that appeared to have the most uniform velocity profile, and hence it was selected for manufacture. The velocity profiles (both of numerical and measurements) at this plane are shown in Fig. 10. Comparison of the uniform flow through the working section, between predicted and experimental results shows close agreement. However, it is noticed that there is small discrepancy (of order $\pm 2\%$.) between the predicted results and measurements.

It was found that of the model tested experienced no separation. In Fig. 11, a typical wall shear plot for model demonstrates the lack of separating flow (indicated by τ_{min}, minimum positive values of wall shear over the entire wall).

The boundary layer characteristics at the exit of the test section are presented in Table 2. The computed boundary layer characteristics from the new design contraction show better agreement with results from experimental wind tunnel tests than the old design based on the analytical technique. It is noticed the improvement on the performance of the wind tunnel as the new design has 38.68% decrease in boundary layer thickness and 28.43% decrease in momentum thickness compared with the old design.

6. Conclusions

In the present contribution, a detailed parametric study is concerned with design optimization technique with (RANS) analysis of flow for design shape of three dimensional geometry of existing open circuit wind tunnel contraction. The present study achieves the recommended contraction ratio, maximum uniformity at the working section mid-plane, prevention of separation, without Gortler vortices in the contraction, and minimizing the boundary layer thickness at entrance to the working section.

The use of CFD has increased the flexibility of shape considered, and allowed the use of design exploration algorithm to optimize the profile of the contraction in an automated manner. The different design points were rated based on a goal-driven optimization using screening method to choose

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the best design set and verified by the CFD solver to define the profile.

It was found that the best result, producing the most uniform velocity profile at inlet to the working section, and preventing separation of the flow within the contraction, was obtained when the point of inflection was located as far downstream as possible.

The extensive numerical simulation illustrates that the hexahedral mesh is more stable and robust to converge. In spite of good results, usage of this type makes the mesh not fine enough near the wall to resolve the characteristics of the boundary layer.

The boundary layer characteristics based on optimization technique show significant improvement on the performance of the wind tunnel.

Physical calibration of the facility has validated the CFD method used and demonstrated that the technique can be addressed for future wind tunnel designs.

The entire study took less than a week compared to few weeks which would be required in the absence of well integrated optimization and CFD tools.

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Ahmed Sharaf Abdelhamed Professor, Mechanical Power Engineering Dept., Faculty of Engineering, Port-Said University, Egypt.



Yassen El-Sayed Yassen Lecturer, Mechanical Power Engineering Dept., Faculty of Engineering, Port-Said University, Egypt.



Mohamed Metwaly ElSakka Demonstrator, Mechanical Power Engineering Dept., Faculty of Engineering, Port-Said University, Egypt.